

To what extent will AI be positive for the UK up until the year 2050?

Introduction

Although it has been 70 years since the first artificial-intelligence based program was written, the growth of artificial intelligence has only become exponential within the western world in the last 5 years or so. Infact, the global A.I market size is expected to grow at a compound annual growth rate (CAGR) of 42.2% from 2020-2027. (Grandviewresearch) Whilst many people are skeptical regarding the ergonomics of artificial intelligence, its gargantuan presence in the future is inevitable, and will revolve a key part around our lives going forward.

As countries across the globe, particularly in Europe, North America and Eastern Asia continue to integrate A.I into their societies, the social and economic impact of this revolution becomes particularly pertinent. Using the case study of the UK, this essay will explore to what extent will artificial intelligence benefit a country in terms of its economy and society.

Purpose of my essay (The debate of A.I growth in the UK)

Artificial intelligence is a very controversial topic within the UK: on one hand, one may believe that A.I will provide major success in improving the economy as well as improving the efficiency of healthcare and research progression. However, others may think that A.I implementation in the UK will not be sustainable. A recent survey conducted by Education Technology recorded that 61% were concerned about the idea of A.I systems. The 2 main reasons due to this mindset is that A.I machines 'lack human intuition and emotion', and that Artificial Intelligence could impact future job security within the UK. (Owen, 2020)

Whilst I fundamentally acknowledge and agree with the thesis that Artificial Intelligence will positively shape up the future of the U.K, I found it very eye-opening of the fact that a lot of people are genuinely worried about the concept of it. This inspired me to conduct my own investigation into other potential risks and dangers surrounding A.I going forward in the UK, as well as reinforcing the perks A.I will provide to the UK.

Structure of the report

The first section of the paper provides an overview of the potential A.I-enhanced factors leading to the inevitable growth in the UK economy, as well as the economic impact artificial intelligence will have partaken in the UK. This 'impact' can be evaluated into 2 segments: the benefits and drawbacks A.I will have on the UK economy. This research devotes significant attention to the UK's GDP as a way of measuring the economic success. I will then use this data as well as the opinions of experts to perform a calculated prediction as to whether artificial intelligence will be beneficial for the UK economy for the next 30 or so years.

The second section of this report delves into the potential repercussions of the UK society in the future as a result of A.I, primarily by observing the jobs created and destroyed due to A.I augmentation. I will then discuss the deeper meanings of how job creation/displacement due to A.I will be positive/negative for society, before providing my overall thoughts stating whether or not artificial intelligence will end up proving merit for society in the UK by 2050.

Artificial intelligence on the UK's economy

A.I investment in the UK in the past

In order to obtain a good understanding of the potential impacts A.I will have on the economy of the UK in 20 or so years, we first need to evaluate previous activity the UK has been involved in regarding artificial intelligence. Jerome Piseni, the author of a recent government report 'GROWING THE ARTIFICIAL INTELLIGENCE INDUSTRY IN THE UK', emphasised that whilst the UK is heavily involved in A.I investment, even trumping other superpowers such as France and Germany in terms of money spent on A.I, still lagged behind China and the US in terms of Artificial Intelligence investment. In fact, the US has invested over £1.6 billion in A.I in 2016, whereas the UK has only managed to spend £150 million within the same year.

However, one promising sign that suggests the UK is becoming more aware of the potential of artificial intelligence is that the vast majority of its investments has come within recent years. For example, in 2011, only \$6 million was invested by the UK, a minimal amount compared to 5 years later, where this tally was \$152 million. (GOV.UK, 2017, pp. 39-40) Furthermore, the UK's GDP has risen by approximately £17 billion, solely from Artificial Intelligence based products. This accounts for almost 7% of the UK's total GDP increase. (PricewaterhouseCoopers, June 2017, p. 12) We can conclude from this that **a positive correlation between the amount of capital invested into A.I based designs, and the revenue earned due to A.I in the UK, does exist.**

Components of A.I that will lead to the increase in the UK's GDP

Given the current trend: that the UK is increasing its resources and investment in the Artificial Intelligence sector, I can confidently predict that **the UK's GDP will increase due to A.I for the next 30 years or so, and I also predict that this increase will accelerate over time as machine learning further develops and evolves.** Infact, current trends suggest that the UK GDP could be up to 10.3% higher in 2030 as a result of A.I. (PricewaterhouseCoopers, June 2017, p. 4) What I really want to focus on is why? There are 2 main segments boosted from A.I augmentation that will lead to this influx in the economy: Productivity and consumer experience.

Productivity: Productivity can be defined as the ratio in business output to business input. Approximately 1.9% of the increase in the UK economy in 2030 will be impacted in consequence to the increased productivity in businesses, largely assisted with artificial intelligence. One application of A.I in boosting productivity, emphasised by Javier Jimenez, is that machine learning advancements has the potential to undertake specific physical procedures instead of human labour in the form of automation. For instance, the physical inventory process, which is a vital procedure for businesses as it's used to calculate their Costs Of Goods Sold. (COGS) Jimenez elaborates that *'AI can also introduce efficiencies with something as simple as taking physical inventory. A task that takes employees one month to complete at Walmart can be completed in 24 hours using sophisticated drones that fly through the warehouse, scan items, and check for misplaced items.'* (Jimenez, 2018)

Business modelling and strategizing can also be guided with A.I systems. A case study demonstrating exquisite A.I utilization in business planning is in Peloton. John Foley, the CEO *'credits the magic of today's technology, including software, data, and communication networks, for the basis of Peloton's success.'* (Gerderman, 2020) Other firms including some in the UK (e.g HSBC installing a new pricing chat-bot for its clients) are beginning to follow suit of Peloton's computational-based model, clearly indicating an upward trend in trust of A.I analytics influencing resource use within businesses.

Both respective applications of A.I mentioned increase productivity within firms by lowering human expenditure in the case of automation, or by maximising optimization of business spending in order to maximise output by implementing Artificial intelligence in business analytics. Higher productivity in the UK increases wages and the overall standard of living, so its economy will thrive in the future as a result of artificial intelligence.

Consumer experience: Firms in the UK will also have the potential to use artificial intelligence in order to enhance consumer products and services. PricewaterhouseCoopers in a report discussing the macroeconomic impact of A.I declare that '*The ability to collect, store and analyse data at the scale, **speed**[1] and in the ways facilitated by AI technologies will allow firms to improve the quality of products and **tailor them to consumers**[2], increasing their inherent value.*' (PricewaterhouseCoopers, February 2018, p. 11) This statement can be dissected in order to represent 2 methods artificial intelligence permits businesses to create better encounters with their customers: 1. Save time for customers by using data mining/analysis tools to more effectively offer products that would likely please them. 2. Applying machine learning approaches in order to acquire a better understanding for customer needs, which can enhance product personalisation for the consumer.

Better customer service (resulted from A.I augmentation) will obviously lead to additional consumption of business goods/services from these customers, so total revenue earned from firms and thus corporation tax towards the economy rises. By 2030, impacts from consumer-experience enhancement due to A.I will approximately comprise 8.4% of the rise in UK GDP. (PricewaterhouseCoopers, June 2017, pp. 11-12)

The regional impacts of A.I in the UK: Widening the 'north-south divide'

It comes as no surprise that England is the province within the UK that will have contributed the most towards the UK economy as a result of artificial intelligence, as it is by far the region of the UK with the highest population. England alone by 2030 would have generated \$205 billion due to A.I, providing over 87% to the total A.I revenue for the rest of the UK.

What really surprised me though is if we zoom in further within England and observe the regions where a lot of A.I related activity is happening, and compare them to places where the case is otherwise. I quickly concluded one disparity: **there is a lot more A.I related activity towards the south of England than at the north.** One approach to measure A.I movement in different regions is by discovering the number of A.I based corporations situated in the given area, and the result is that the vast majority of these corporations are in Southern England. London is the strongest centre of A.I startups, with 80% of them being located at the capital. Other southern cities such as Oxford, Cambridge and Bristol also have a large presence in A.I companies, with Bristol in particular housing tech firms like HP and Oracle working in artificial intelligence. (GOV.UK, 2017, p. 30)

Why is this discrepancy of A.I development between the north and south significant? The reason is at the time of writing, there is already a sense of inequality between the north of Leicester being considerably poorer than the rest of England, more commonly known as the 'North-South divide'. Katie Burton in an article for Geographical expressed her concerns for this as she believes that '*Even after adjusting mortality rates for deprivation, a substantial divide remains, suggesting more **deep-seated structural issues***'. (Burton, December 2020) Without careful supervision, the future integration of A.I into the economy could widen the UK's regional inequality gap.

Resolving the North-South divide of A.I : My proposed solution is relatively simple: the UK government should encourage technology related companies to expand their operations towards the Northern areas of the UK in order to bridge the development gap of artificial intelligence between the north and south. One approach is to reduce the corporation tax in northern cities such as Liverpool and Manchester, which can incentivize A.I firms to move their quarters into these regions as their running costs will decrease. While this may lower government income from the technology sector for the first 5 years doing this will certainly provide balance when it comes to the geographical spread of Artificial Intelligence in the UK. Once the presence of A.I corporations in the north is sufficient enough, corporation tax value should be reverted to its original state.

Verdict

I found Burton's remark regarding the North-South divide being a '*structural issue*' to be eye-opening in terms of understanding the true severity of the situation. With the future augmentation of A.I in the richer areas of the south, I fear that northern England will lack the technology development that it requires, and that the UK as a nation will be divided up even further. Therefore, I firmly believe that **the government's handling of the North-South divide using A.I will play a fair part in dictating the economic success A.I will possess in the UK in the near future.**

With that being said, for me, PwC's data regarding the expected GDP increase of the UK due to A.I, combined with Piseni's information of how the UK government have positively handled with A.I in recent times, trumps Burton's reason for concern, and has convinced me to conclude that **there are great opportunities for AI to help the economy of the UK, and that the government are competent enough to realise these opportunities.**

Although the boost in productivity from A.I will play a significant role in boosting the UK economy, I would argue that **A.I intervention in enhancing customer experience will be the dominant factor for the project economy rise, especially towards the rear-end of 2050.** The fact that it will take some time for businesses to produce supply A.I-upgraded products for customers suggests to me that the UK economy boost from A.I initially be driven by productivity-related impacts, whereas after this phase consumer-side impacts will dominate as previously suggested.

The impact of Artificial Intelligence on society in the UK

'The future state of any single job lies in the answer to a simple question: To what extent is that job reducible to frequent, high-volume tasks, and to what extent does it involve tackling novel situations?' (Goldbloom, August 2016)

How A.I has the power to create and destroy jobs in the UK

As the development of artificial intelligence schemes naturally improves over time, there will be a point in which many jobs can be completed more efficiently and cost-effectively through robots than through humans. When firms approach this scenario, the corporation then has a choice to make: retain or discharge its employees? Most companies will likely replace these workers with robots, which is known as the displacement effect.

However, there is a good reason why they may hold on to these workers. Instead, A.I firms could reduce the prices of their products (as the robots have saved the company running costs). As a result, more customers are likely to be interested in purchasing the product they are selling, so the existing workers can stay in order to produce the influx of goods demanded. Furthermore, more workers can be hired if the demand of goods is very high. Technological firms will consider the future demand of their products to decide whether or not they require the additional manpower of the existing employees.

Zooming in on the jobs that will benefit due to A.I in the UK

One common misconception I notice against the idea of automation and other A.I based programs is that it will take over all of the UK population's jobs. However, this could not be any further from the truth. By 2037 20.2% of jobs will be created due to Artificial Intelligence, which equates to 7.2 million jobs in the UK. Below are some of the sectors that will experience the greatest net job boost from A.I. (PricewaterhouseCoopers, July 2018, p. 41)

Healthcare: I estimate that the health and social care sector will experience the greatest net job gain in the UK due to A.I within the next 30 or so years: by 2037: A.I will likely have created almost 1 million jobs in the NHS. This is because society of course values their health over anything else, and as the UK population gets older, healthcare availability becomes increasingly in demand. Furthermore, due to the average income per person projected to increase over the next decade, more people will have the capability to get access to the healthcare system. Due to this influx in demand, more workers will have to be hired. This is where A.I comes in: Artificial intelligence and robotics will be efficient enough to enhance the doctors and nurses capabilities in their work, but not too effective so that these machines will displace the staff. After all, *'anything that requires skills like empathy or interpersonal skills, those things are going to be human for a very long time'*. (Muchhala, July 2019) Healthcare is a prime example of a sector that represents these very virtues (nursing and doctoring occupations require a strong 'human-feel' towards its patients) so these cannot and will not be overrun by robotics for a *'very long time'* in the UK, absolutely much after 2050.

Education: I would also put teaching in the list of sectors 'untouchable' from A.I in the UK. This is due to the fundamental reason that teaching requires high levels of interpersonal skills that are natural in humans, but is difficult to replicate in the form of a robotic system, so these jobs will be available for a *'very long time'* like Muchhala mentioned previously. Furthermore, I strongly believe that A.I has the potential to complement very well with teachers. For example, human teachers can specialise in teaching the curriculum of work whereas A.I machines can contribute to more routine tasks such as marking homework and tests. By 2037, approximately 187,000 jobs will be created in the education sector solely due to A.I.

Research: In a recent blog written by Professor Andrew Briggs (works for the University of Oxford), he was extremely optimistic about how Artificial Intelligence will enable new scientific discoveries. The point he hammered home was that *'by 2025, AI will be as ubiquitous for running experiments as computers are today for controlling instrumentation and logging data. The paradigm shift will be from AI used for analysing the data which has already been obtained, to AI deciding what to measure next. The key advances in AI that we are currently exploring to make this possible are reinforcement learning and Bayesian optimization.'* (Briggs, November 2017) This is just one example that suggests **scientists are delighted with the presence of A.I:** more evidence to support this claim is that PwC predict that in 2037, scientific-related jobs will increase by 16% due to A.I in the UK, which conveys that a net amount of people are moving into research in hopes that artificial intelligence will accelerate the boost of discoveries in the future.

A.I development and management: Of course, Artificial Intelligence in the UK is still in its early stages of development in comparison to other superpowers, so there will be a vast number of jobs available that consist of evolving, testing and implementing artificial Intelligence based programs into the real-world. The quantity of these jobs created is likely to accelerate over the decade as A.I becomes more prominent and widespread in the UK, and the lucrative salary it provides is likely to attract much of the population. According to the World Economic Forum, there will be 58 million new jobs in A.I worldwide by 2022. Given the fact that the UK has approximately a 4% share in the global A.I investment scheme (The Future of Jobs Report 2018, 2018), I estimate that **2.3 million new jobs directly relating to Artificial Intelligence will be available in the UK by 2022.**

Zooming in on jobs that will be harmed by A.I in the UK

Artificial Intelligence will likely displace a very similar number of jobs in the future in the UK than it will create: approximately 7.01 million jobs by 2037. (PricewaterhouseCoopers, July 2018, p. 41) Below are some of the sectors that will experience the greatest net job deficit as a result of A.I.

Manufacturing: I estimate that Artificial Intelligence will detriment the manufacturing sector the most in terms of the role of human activity over the next 30 years. By 2037, there could potentially be 25% fewer jobs in the manufacturing sector due to the augmentation of automation, which equates to a net 680,000 job loss in the UK. I conclude that there are 2 main reasons leading to this displacement: the first is that automated machines are much more productive than human staff. As machinery has the capability to operate during the night as well as the day, they have the potential to produce twice as much output as staff. The other cause of displacement is that automation is more cost-effective than staff: as wages do not have to be paid for machines, corporations can save their running costs by reducing the number of jobs in their workforce.

Rory Cellan-Jones (a technology correspondent) is one of the people concerned of a potential job redundancy in manufacturing as a result of automation. He stresses that *'Oxford Economics also found the more repetitive the job, the greater the risk of its being wiped out.'* (Jones, June 2019) Taking this detail into account, I predict that whilst Artificial Intelligence will not have evolved enough to perform complex tasks, intricate product manufacturing (producing more personalised and high-end products etc.) will likely not experience much of a drop in human labour, whereas **batch and mass production-schemes (that are significantly more repetitive) are considerably more susceptible to displacement via automation in the UK** because like Goldbroom stated earlier, these are frequent and high-volume tasks.

Transportation: Due to the introduction and expansion of self-driving vehicles in the UK, jobs in the transport sector such as taxi and train drivers will eventually become redundant. This is because theoretically, automated vehicles should perform much safer, so economic factors for the firm owning the transport service such as insurance will not be as significant and costly. By 2037, there will likely be 690,000 fewer jobs in the transport industry due to A.I in the UK, which is actually a figure much less than I anticipated. I have come to realise that the only explanation for this is

that even by the next 30 years, the concept of an A.I application to create self-driving vehicles would have still been incomplete and faulty, so the way of transport wouldn't have been fully revolutionized in some rural areas of the UK. Perhaps the development of self-driving vehicles would be complete by 2070, and the transport-related jobs would have plummeted by this time, but considering this will have violated the time range I have given myself for my enquiry, so this hypothesis is invalid.

Retailing: With the rapid augmentation of self-checkout machines in the UK supermarkets, and even some supermarkets such as Sainsbury's announcing further self-service schemes such as the 'Sainsbury's Smartpay', it is fair to suggest that **the emergence of Artificial Intelligence may have contributed to the fall of cashiering jobs in the UK**. My thoughts on the cause of this displacement are heavily similar to the reasonings I gave for manufacturing: automation is simply becoming increasingly more cost-efficient than human labour. Even Cellan-Jones's discussed-earlier theory of repetitive jobs being increasingly displaced can be applied to instances such as cashiering, which explains why by 2037 A.I could displace up to 3% more jobs in the wholesale and retail sector than it would have created. (All numerical figures not already cited until this point from PricewaterhouseCoopers, July 2018, pp. 42-44)

My thoughts on the future effects of Job creation and displacement due to A.I in the UK

There isn't much point declaring that there will be an influx in job creation in education due to A.I, without explaining the potential repercussions of this on society. What are the drawbacks of job displacement? Are there perhaps any potential benefits that may result from job displacement?

Job creation: As A.I produces more and more jobs in the UK, I predict that the unemployment rate in the education, healthcare and research sectors will decrease, as the supply of jobs in these industries will be greater than the demand for these very jobs. If we observe periods of time in the past globally (and thus also in the UK) where unemployment rates were low, I observed that the average wage increased (Economist's View, 2014) (arguably as a method of retention by the employers, as there were a very limited number of alternative workers to hire) so in general, the population had more disposable income. Linking this hypothesis back to the presence of Artificial Intelligence in the UK, I forecast that **workers in education, healthcare and research will seek a better quality of life than people in other occupations in the UK as a result of an increase in job availability created from A.I applications**.

However, in some sectors, Artificial Intelligence may have created too many jobs for its own good. I am going to use the UK healthcare industry (post-introduction to A.I) to represent this potential ambivalence of having too many jobs available. We have already established that there will be more jobs in the National Health Service due to augmentation of A.I within the next 20 or so years, but this fact combined with the fact that there are less people available to employ (from low unemployment) could potentially have catastrophic effects. If there are less people to employ, the NHS may have no option but to hire newly qualified graduates. Josh Bersin (who specialises in human resources tech) disagrees with this approach: he passionately states that *'college graduates are not bringing enough technical or soft skills to the job market (only 35% of employers feel grads are sufficiently ready) and four in ten are building remedial programs to try to fill this gap. So while there are a lot of jobs open, we are seeing an increasingly wide gap between the jobs being created and the skills and experiences in the workforce to fill them.'* (Bersin, July 2018) If we apply Bersin's pessimism of graduates into our microcosm in the NHS, the worst case scenario I believe could happen due to an uncontrollable quantity of job creation from A.I is that the NHS are obliged to hire inexperienced and incompetent graduate students from colleges and universities, which will likely result in a loss in productivity. In a sector as crucial and sensitive as the healthcare sector, I fear that a decrease in job productivity may lead to lives being lost.

Job displacement: Theoretically, the impacts of job displacement due to Artificial Intelligence in the UK is in reality just the inverse of what we discussed about job creation: Unemployment rates in the transport, retail and manufacturing industries increase, firms have more hirees available at their disposal, so the average wage in these sectors will likely decrease.

What I really want to delve into however is the mental fatigue caused by job displacement due to A.I, which I believe is a truly underappreciated topic that is not discussed or acknowledged enough during my research. Tonya Riley (a researcher interested in A.I) expressed concern regarding A.I augmentation around the world, as she believes that *'workers in countries facing higher levels of automation risk reported more frequent levels of physical and mental distress, as self-reported using a rubric provided by the County Health Rankings. The counties predicted to be most affected by automation also appeared lower in a broader health ranking, which was based on factors including lifespan, alcohol and drug use.'* (Riley, May 2018) The detail that struck me the most is the 'lower lifespan' impact as a result of job displacement via automation. Taking this article into account, I predict that the **mental issues created by job displacement due to A.I have the potential to affect the UK population more severely than any other issues, such as finance.**

Verdict

Although A.I is notoriously perceived in society for displacing our jobs, figures outline that **there will be a decisive net creation in jobs in the UK in the future as a result of artificial intelligence.** Although I think that Cellan-Jones and Riley's apprehensions regarding A.I on society are valid and should be considered by administrators, I personally relate my thoughts more towards Briggs optimism because an influx in research job interest from A.I utility can help humanity discover great things, especially within the time frame of 2050.

I believe that the handling of artificial intelligence industrialisation by the UK government will play a pivotal role regarding how A.I intervention in society is perceived by experts and researchers in the future, especially in their success in acknowledging and addressing the risk of mass job displacement in specific sectors due to A.I in the future. Detailed research needs to be conducted in order to identify the specific jobs at risk of job displacement via automation (which of course primarily will be manufacturing, transportation and retailing related occupations), and appropriate interventions need to take place: *'Such interventions could include supporting businesses to retrain employees, and providing financial and psychological support to the people impacted.'* (Fenech, April 2018) I agree that both of these impacts mentioned by Fenech would successfully address the impacts of job displacement.

Overall, I DO think that the UK government has the capabilities to apply these interventions to minimize the risk of artificial intelligence on society, similarly to my reasoning before that they exhibited great handling of A.I in terms of investment in the past, as discussed previously. If these proceedings do go as planned, I confidently project that **by 2050 UK administrators will have gained enough experience to maximise artificial intelligence utilization within our society.**

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